

CS671 – Deep Learning and Its Applications

Technical Documentation Report

Model: Paired Diffusion (Brain)

Project Group: 5 Region: Brain Modality: Paired (CT → MRI)

1. ARCHITECTURAL OVERVIEW

Paired Diffusion utilizes a Latent Diffusion Model (LDM) framework to perform CT-to-MRI synthesis. Unlike GAN-based approaches, diffusion models learn the data distribution by reversing a gradual denoising process. In this implementation, the model is conditioned on CT features to guide the generation of aligned MRI slices in a compressed latent space.

1.1 Latent Diffusion Architecture

The system consists of three primary components: a Variational Autoencoder (VAE), a Conditional UNet, and a conditioning encoder.

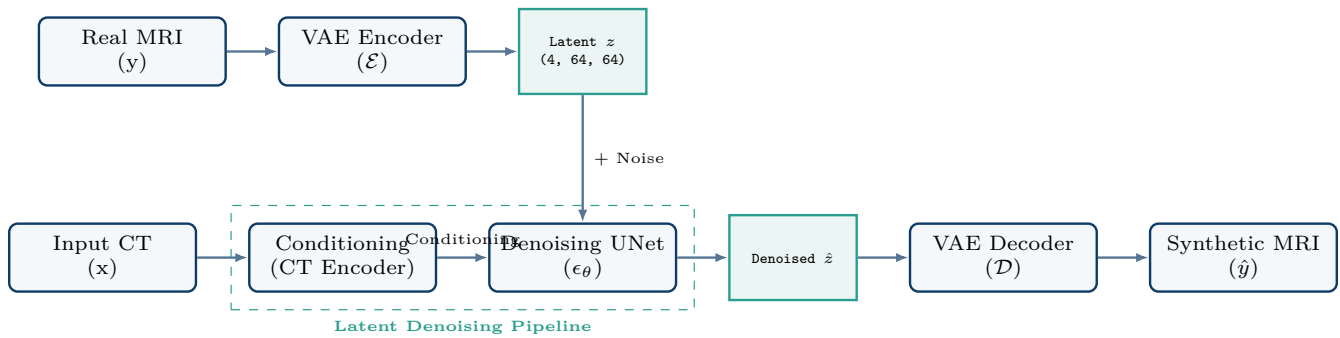


Figure 1: Latent Diffusion Architecture: Denoising in the VAE latent space conditioned on CT features.

VAE Latent Space: To reduce computational complexity, the model operates on a $64 \times 64 \times 4$ latent representation instead of the 256×256 pixel space. The VAE is pre-trained to compress medical images while preserving high-frequency anatomical features.

Conditioning Mechanism (CT Control): The input CT slice is processed by a conditioning encoder (often a ResNet or a frozen CLIP-like encoder) to extract spatial and semantic features. These features are injected into the UNet via Cross-Attention layers, ensuring that the denoising process "sees" the CT anatomy at every scale.

Denoising UNet: The core of the model is a UNet with self-attention and cross-attention blocks. It learns to predict the noise ϵ added to a latent z , given the current timestep t and conditioning x .

2. TRAINING WORKFLOW

Training involves a forward diffusion process (adding noise) and a learned reverse process (denoising).

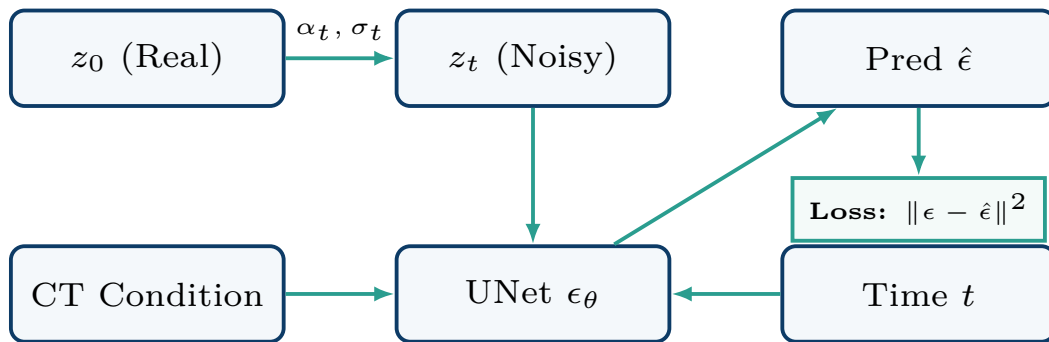


Figure 2: Forward and Reverse Diffusion: Learning to denoise conditioned on CT and Time.

Objective Function: The model minimizes the Mean Squared Error (MSE) between the actual noise ϵ and the predicted noise $\hat{\epsilon}$. By training across all timesteps $t \in [0, T]$, the model learns to reconstruct the MRI manifold from pure Gaussian noise, guided by the CT conditioning.

3. INFERENCE PIPELINE

Inference (sampling) is an iterative process where the model starts from random noise and gradually removes it to reveal the synthetic MRI.

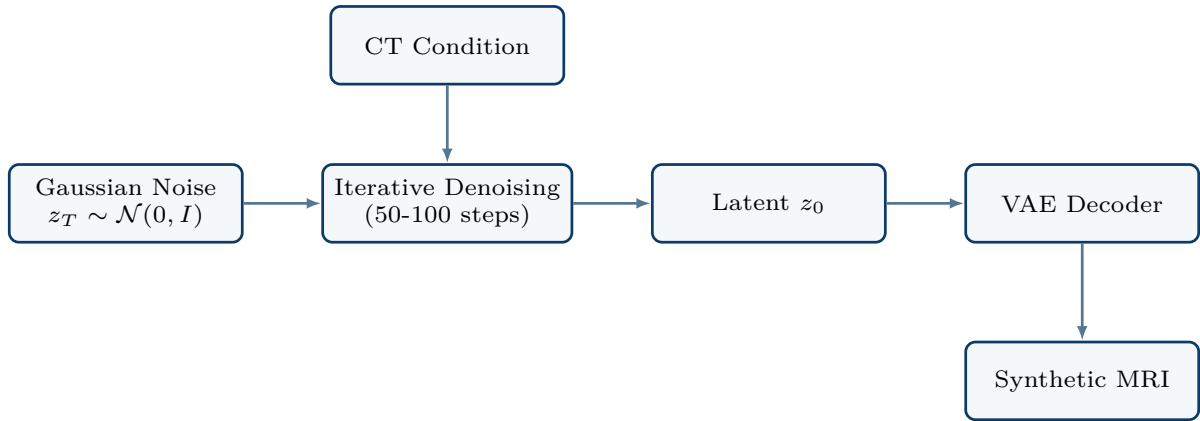


Figure 3: Inference Sampling: Iterative reconstruction guided by CT conditioning.

During inference, we use schedulers like DDIM or PNDM to accelerate the sampling process. The input CT volume is encoded once, and its features are used as constant conditioning throughout the denoising iterations. This ensures that every generated MRI slice is perfectly aligned with the anatomical structures present in the CT.